THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SOCIAL SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Minerals and Energy Resources (Geography)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A village installs a cow dung digester for cooking gas. This is 1

(a) biogas **(b)** CNG

(c) geothermal energy **(d)** nuclear energy

2. Which pair is INCORRECT? 1

(a) Jharia - coal **(b)** Tarapur - nuclear power

(c) Mumbai High - coal **(d)** Panchpatmali - bauxite

3. Which statement is correct? 1

(a) non-conventional energy sources are eco-friendly

(b) coal is renewable

(c) natural gas is solid

(d) oil is a biotic resource

4. Which statement about rat-hole mining is correct? 1

(a) it is large scale **(b)** it is offshore

(c) it is dangerous **(d)** it is regulated and safe

5. Which of the following is NOT a use of mica? 1

(a) paints **(b)** electrical insulators

(c) electronic goods **(d)** making steel

6. Assertion (A): Rat-hole mining raises safety concerns. Reason (R): It uses narrow tunnels that can collapse or flood. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

7. Rising international oil prices would most affect 1

(a) India's tea exports **(b)** India's import bill

(c) India's forest cover **(d)** India's monsoon

8. Assertion (A): Coal is the most abundantly available fossil fuel in India. Reason (R): It is used for power generation and industry. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. **Assertion (A):** Lignite is a low-grade brown coal that is soft with high moisture content. 1
Reason (R): Lignite has the highest carbon content of all types of coal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Hydroelectricity is a non-renewable source of energy. 1
Reason (R): Hydel power is generated from the flow of water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why has CNG been introduced in cities? 2

12. Why is solar energy hard to scale? 2

13. Why must mining firms restore land? 2

14. Why do mineral exports raise concern? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Analyse why India depends on imported crude oil. 3

16. Explain why renewable energy needs research and investment. 3

17. Analyse why oil refineries are located near ports. 3

OR

Explain how technology improves mineral use. 3

18. Explain why coal-based power is criticised. 3

OR

Explain how India can reduce fossil fuel dependence. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a village in Uttar Pradesh, a farmer set up a plant that uses cattle dung and farm waste to produce gas for cooking. Decomposition of organic matter yields gas which has higher thermal efficiency than kerosene, dung cake and charcoal. The plant also gives improved quality manure for his fields. Plants using cattle dung are known as 'Gobar gas plants'. 5
(i) What is the energy resource produced in the plant?
(ii) Is it conventional or non-conventional?
(iii) Name any one of its two benefits to the farmer.
(iv) Why is biogas more suitable for rural India than dung cakes or firewood?

20. Compare conventional and non-conventional energy resources with examples. 5

OR

Evaluate how citizens can contribute to conserving energy and minerals. 5

END OF QUESTION PAPER — Total: 40 Marks